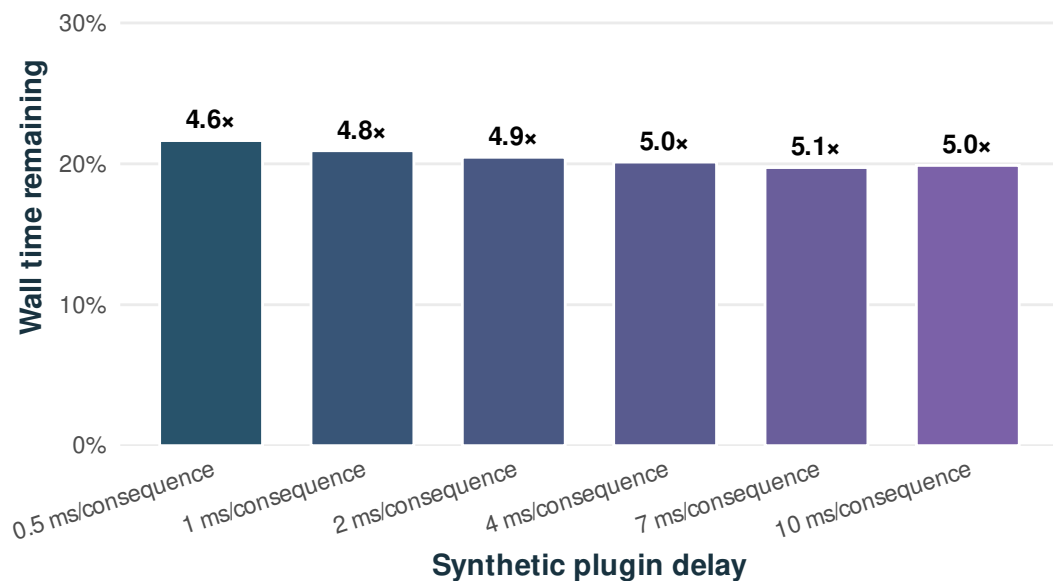


VCFcache converts annotation complexity into time saved

One 4.80-million-variant real WGS · bundled Zenodo gnomAD AF $\geq 10\%$ cache · ~80% hits · light statistics · no technical repeats

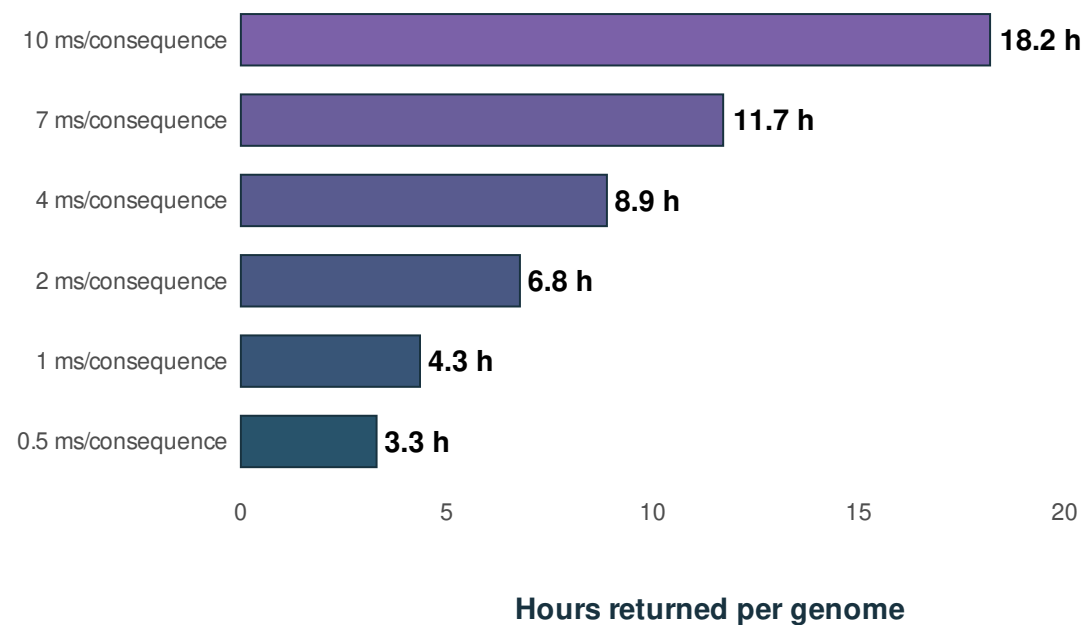
A About 80% reuse removes most of every pipeline

Numbers above bars are measured speedups



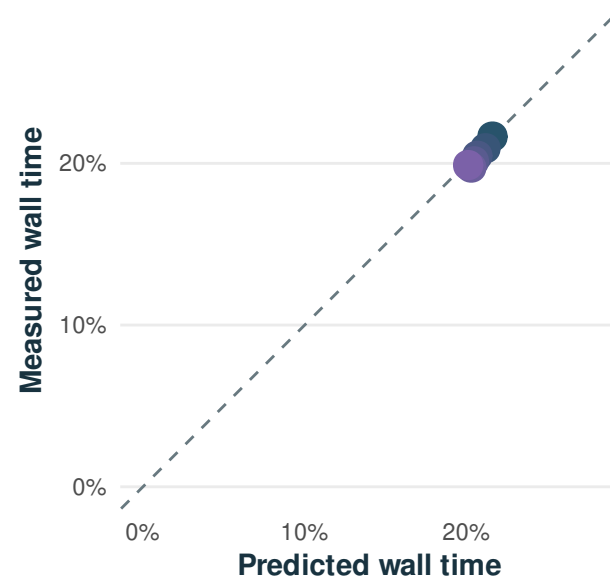
B Slower pipelines return hours, not minutes

The same real WGS and cache for every load



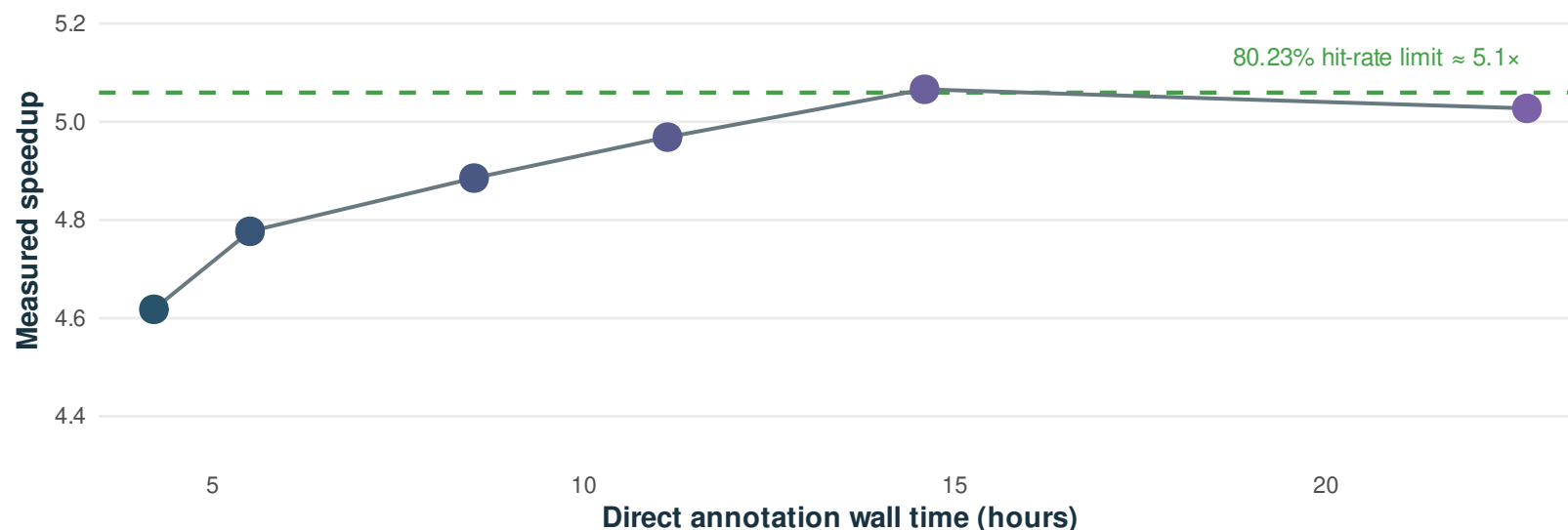
C The simple miss-fraction model remains predictive

Tcached = measured VEP lookup overhead + (1 - hit-rate) * time returned



D Speedup approaches the hit-rate ceiling as pipelines slow

As pipeline slows, time returned keeps growing after relative speedup saturates



All six cached outputs were semantically equivalent to their pipeline-specific uncached output. Synthetic delays run only for transcript consequences submitted to VEP and emit no annotation fields.